

**IMSE 586: Big Data Analytics and Visualization**

**Fall 2024**

**Project Proposal**

**Tentative title: Predicting Stock Market Performance Using Historical Price  
Data and Technical Indicators**

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## Background

The stock market is a volatile and complex market, influenced by many different factors. Stock price prediction is a very valuable skill that allows investors, financial analysts, and data scientists to make data-driven decisions. Using historical data on the stock market, this project examines some of the patterns and factors responsible for the movement in stocks, with particular emphasis on modeling techniques in forecasting stock performance. These findings could help investors predict some trends that might happen in the future and understand the levels of risk.

## Research Questions

1. What indicators do you think are most responsible for stock price movement with your chosen dataset?
2. How accurately can you predict the closing price of the stock based on the historical data and technical analysis?
3. What patterns in volume and price changes can be identified that correspond to high or low performance periods?

## Literature Review

1. **Efficient Capital Markets: A Review of Theory and Empirical Work** by E.F. Fama (1970). This project introduces Efficient Market Hypothesis, which asserts that "stock prices reflect all available information." While the hypothesis has been widely debated, it certainly underlines the fact that, indeed, it is tough to predict market movement. This project looks at practical predictability of stock prices using historical and technical indicators.[1]
2. **A prediction-based clustering algorithm for stock market trading** by P.E. Tsinaslanidis and D. Kugiumtzis (2017). This research applies some clustering techniques to categorize behaviors in stock prices that can be of assistance in finding similar patterns for future predictions. The Work analyzes the price fluctuations and some technical indicators for an enlightening approach on how machine learning can be applied to predict market movements.[2]

## Data Set(s)

**Primary Dataset:** Yahoo Finance Stock Market Data - **Source:** User-provided file containing daily historical data with columns for Date, Open, Close, High, Low, Volume, and Adjusted Close. - **Description:** This dataset spans daily stock prices, enabling exploration of price trends, volume fluctuations, and prediction of future values. - **URL:** <https://www.kaggle.com/datasets/arashnic/time-series-forecasting-with-yahoo-stock-pr>

## Method Description

The following analysis and modeling methods will be used for the development of answers to the research questions:

1. **Data Preprocessing and Exploration:** Data will be cleaned and preprocessed using pandas, handling missing values, removing outliers, and feature engineering on technical indicators such as moving averages and the relative strength index.

2. **Exploratory Data Analysis (EDA):** We will visualize stock trends over time by using matplotlib and seaborn, putting the spotlight on the price-volume-volatility relationship. We shall also plot the distribution of daily returns to understand the variability in the performance of the stocks.
3. **Feature Engineering:** We are going to compute moving averages, exponential moving averages, and indicators based on volume. These will help in building a strong feature set that would be useful in a predictive model.
4. **Modeling and Prediction:**
  - Linear Regression: Building the baseline model to predict the stock closing price with the help of past values.
  - Logistic Regression: To classify high and low performance based on the percentage variation in price.
  - Time Series Analysis: Application of rolling statistics to analyze the trend and seasonality.
  - Cross-Validation: Techniques like k-fold cross-validation will make sure that the models are accurate and not overfitted.
5. **Visualization and Dashboard:**
  - Time series performance of the stocks to be visualized, both statically and interactively, in matplotlib and Plotly.
  - An interactive dashboard showing the trends and indicators will be prepared using Jupyter Notebook.

## Preliminary Progress

- A preliminary view of the dataset seems to indicate that it contains data columns like Date, Open, High, Low, Close, Volume, and Adjusted Close, and thus is a good starting point for the analysis of stock prices.
- Key indicators have been identified for feature engineering and model development.

## References

- [1] Eugene F. Fama. “Efficient Capital Markets: A Review of Theory and Empirical Work”. In: *The Journal of Finance* 25.2 (1970), pp. 383–417.
- [2] P. E. Tsinaslanidis and D. Kugiumtzis. “A prediction-based clustering algorithm for stock market trading”. In: *Expert Systems with Applications* 80 (2017), pp. 115–123.